

Step 2 Validation Protocol: Separating Constant SOC, Dimerisation, and Local Spin-Correlation Controls

1 Purpose of the test

The purpose of this step is to determine whether the minimal bright/dark spectroscopy model can distinguish three phenomenological mechanisms for the effective bright-dark mixing:

$$V_{BD}^{\text{eff}} = V_0, \quad (1)$$

$$V_{BD}^{\text{eff}} = V_0 + \lambda_\delta \delta, \quad (2)$$

$$V_{BD}^{\text{eff}} = V_0 + \lambda_C C_1. \quad (3)$$

The test is not a microscopic calculation of CuGeO₃. It is a controlled spectroscopic check built on the Step 1 model.

2 Physical model

The Hilbert space contains a ground state, a dark excited state, and a bright excited state. The bright state carries the optical dipole by default. The dark state becomes visible only through bright-dark mixing. The effective mixing is

$$V_{BD}^{\text{eff}} = V_0 + \lambda_\delta \delta + \lambda_C C_1. \quad (4)$$

Here δ is a spin-Peierls/lattice dimerisation order parameter, while $C_1 = \langle S_i \cdot S_{i+1} \rangle$ is a local spin correlation that can be nonzero even in the uniform chain.

3 Expected results

If V_0 alone is active, the cross-peak signal should reproduce the qualitative Step 1 bright/dark result. If only $\lambda_\delta \delta$ is varied, changes in the cross peaks should track the dimerisation amplitude. If only $\lambda_C C_1$ is varied, cross-peak changes can remain present even when $\delta = 0$, which would distinguish spin-correlation control from a purely spin-Peierls order parameter effect.

4 Numerical procedure

Run one scenario at a time by changing `active_case_key` in the notebook or script. The third-order rephasing 1Q spectrum is generated with the same NQ protocol used in Step 1. Each run writes a spectrum PDF, an NPZ file containing the complex response arrays, and text/JSON summaries.

5 Acceptance criteria

This step is qualitative and comparative. The model is considered useful for Step 2 if the three controls produce distinguishable changes in at least one of: cross-peak area, cross/diagonal ratio, peak position, or pathway-resolved amplitude. Absolute convergence in frequency-grid resolution is not required for this first separation test.

6 Requested figures and data

For each scenario, save the rephasing real spectrum, the pathway-resolved response arrays, the total rephasing component, and the run parameters. Later Phase 2 analysis should compare cross-peak finite-window integrals and cross/diagonal ratios across the three scenarios.

7 Interpretation of possible failures

If all three controls produce nearly identical spectra at matched V_{BD}^{eff} , then the current observable is sensitive only to the total effective mixing, not to its microscopic origin. If the dimerisation and spin-correlation controls differ only through amplitude, additional observables such as temperature/field trends, unrepasing pathways, or dynamic phonon coupling may be required.

8 Numerical results

The three requested scenarios were generated at $\eta = 0.001$ with a 100×100 frequency grid. The finite-window half-width used for the aggregate metrics is 0.025 eV. The reported cross area is the sum of the two cross windows, and the ratio is the cross-window area divided by the sum of the two diagonal-window areas.

Scenario	V_{BD}^{eff}	δ	C_1	$A_{\text{cross}}/A_{\text{diag}}$
Constant SOC	0.0100	0	0	0.120590
Dimerisation control	0.0200	0.0500	0	0.105788
Spin-correlation control	0.0200	0	-0.443	0.105788

The summed cross-window finite-area values are 10.210040 for constant SOC and 13.682378 for both enhanced-mixing controls. The direct spectrum difference between the dimerisation and spin-correlation controls at matched $V_{BD}^{\text{eff}} = 0.02$ has relative maximum difference 2.62×10^{-17} .

9 Validation outcome

The first criterion passes: the model remains sensitive to the effective bright-dark mixing, since the constant-SOC case and the enhanced-mixing cases produce distinguishable cross-window amplitudes.

The microscopic-origin criterion is inconclusive in this scalar model. The dimerisation and spin-correlation controls are numerically identical at matched V_{BD}^{eff} , because δ and C_1 enter the Hamiltonian only through the same scalar effective mixing. Therefore this Step 2 model can separate weak and enhanced bright-dark mixing, but it cannot by itself assign the enhancement to spin-Peierls dimerisation rather than to local spin correlations.

10 Conclusions

This step gives a useful negative control. The nonlinear response is clearly sensitive to the magnitude of the effective bright-dark mixing: increasing V_{BD}^{eff} from 0.01 to 0.02 changes the cross-window signal and the full complex spectrum. This confirms that the Step 1 observable remains usable when the mixing is promoted from a fixed SOC parameter to an effective control parameter.

However, the present implementation cannot distinguish the microscopic origin of the enhancement. In the Hamiltonian used here, δ and C_1 only enter through the single number V_{BD}^{eff} . Consequently, a dimerisation-driven enhancement and a local-spin-correlation-driven enhancement are exactly equivalent when they produce the same V_{BD}^{eff} . The matched comparison gives a relative maximum spectral difference of 2.62×10^{-17} , which is numerical zero for the present calculation.

The physical conclusion is therefore limited but important: a temperature- or field-dependent cross peak cannot be assigned to SOC, dimerisation, phonons, or spin correlations from this scalar-mixing model alone. The model can detect an enhanced bright-dark mixing channel, but it cannot identify whether the enhancement is caused by spin-Peierls lattice order or by correlations already present in the uniform spin chain.

11 Proposed next steps

11.1 Step 3A: add a dynamic dimerisation/phonon coordinate

The most direct next step is to give the dimerisation channel a dynamical signature rather than only a scalar amplitude. A minimal implementation would keep the bright/dark sector but add an explicit phonon coordinate Q :

$$H_{\text{mix}} = (V_0 + \lambda_\delta \delta) (|D\rangle\langle B| + |B\rangle\langle D|) + g_Q Q (|D\rangle\langle B| + |B\rangle\langle D|). \quad (5)$$

This should create phonon sideband structure, modified pathway phases, or additional dephasing/lineshape changes that a static C_1 control does not produce. This is the cleanest route for separating lattice-driven spin-Peierls physics from a purely scalar SOC enhancement.

11.2 Step 3B: add independent temperature and field trends

The second control should be a trend-level comparison:

$$\delta(T, B) \rightarrow 0 \quad \text{above the spin-Peierls transition,} \quad (6)$$

$$C_1(T, B) \neq 0 \quad \text{in both uniform and dimerised regimes.} \quad (7)$$

If a spectral feature follows $\delta(T, B)$ and vanishes above the transition, it is compatible with spin-Peierls/lattice control. If it persists when $\delta = 0$ but C_1 remains finite, it should not be attributed uniquely to the dimerised phase.

11.3 Step 3C: introduce a k- or bond-sector distinction

The third step is to make the dimerised and uniform spin-chain controls act on different sectors. For example, a dimerised phase can fold the Brillouin zone or mix k and $k + \pi$, while a uniform-chain correlation control should preserve the original translation symmetry. This would test whether rephasing and unrephasing spectra are sensitive to a genuine spin-Peierls sector structure, not only to a scalar change in V_{BD}^{eff} .

11.4 Step 3D: compare more observables than cross-peak area

The Step 2 area metric is useful but not sufficient. Future runs should compare at least:

- rephasing versus unrepasing totals;
- real, imaginary, and absolute spectra;
- cross/diagonal ratios;
- pathway-resolved amplitudes;
- peak positions and linewidths;
- possible phonon sidebands or asymmetric lineshapes.

The goal is to find an observable that changes under dimerisation/phonon dynamics but does not change under a purely static C_1 scalar control.

11.5 Recommended order

The recommended next model is Step 3A: keep the same bright/dark model and add one explicit dimerisation/phonon coordinate. This is the smallest change that can break the degeneracy found in Step 2. After that, Step 3B should test whether the resulting spectral signatures follow realistic temperature and field trends for $\delta(T, B)$ and $C_1(T, B)$.